

HIGH ACRYLONITRILE NBR: PATENT LANDSCAPE ON SOLUTION POLYMERIZATION, LATEX BLENDING, AND PRECROSSLINKED POWDER FORMS

1. ABSTRACT

The supplied patent set spans three distinct but related approaches to high-acrylonitrile nitrile-butadiene rubber (NBR) technology. US9023914B2 focuses on solution polymerization of nitrile rubbers in organic solvent using specific modifier substances, with optional post-hydrogenation to HNBR. This patent emphasizes control of molecular architecture, low polydispersity, and incorporation of modifier-derived fragments into the main chain and/or chain ends. The disclosed process is batchwise in a 5 L autoclave and uses free-radical initiation under elevated temperature, with examples showing butadiene metered under pressure and reactions run at about 100 °C for several hours to more than 20 hours, with conversions reported from about 9.2% to 56% depending on example and conditions. US8470422B2 addresses a different problem: tailoring carboxylated nitrile-butadiene latex formulations for glove articles that mimic natural rubber feel. Its key technical contribution is blending two acrylonitrile terpolymers in a 60:40 to 40:60 ratio, with total acrylonitrile content between 17 and 45 wt% and methacrylic acid below 15 wt%, while balancing ionic crosslinking and sulfur/accelerator cure to achieve lower modulus and improved elongation. EP3902855B1 shifts from latex article performance to powder processing, claiming pulverulent mixtures of precrosslinked nitrile rubbers containing acrylonitrile, a conjugated diene, and 0.5–5 wt% trimethylolpropane tri(meth)acrylate, combined with separating agents to improve handling and downstream compounding, especially for tire tread applications. Across the set, the recurring themes are control of acrylonitrile-containing copolymer composition, use of process additives to tune molecular weight or crosslink density, and selection of workup/form factor (solution polymer, latex blend, or powder) to match the intended application. The patents collectively show a progression from synthesis-centric control in solution, to property tuning in latex articles, to powder-form processability and compounding performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015-05-05
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Initiator: 95 ml (initiator ... 95 ml) - Pressure: 71 ml (pressure ... 71 ml) - Pressure: 3 bar (pressure ... 3 bar) - Conversion: 39 % (conversion ... 39 %) - Conversion: 48 % (conversion ... 48 %) - Conversion: 9.2 % (conversion ... 9.2 %) - Time: 22 hours (time ... 22 hours) - Time: 6 hours (time ... 6 hours) - Dmac: 2 g (2 g of Dmac) - Autoclave: 5 L (5 L of Autoclave) - Dispersion: 50 % (50 % of Dispersion) - Fe: 0.986 g (0.986 g of Fe) - Rongalit: 2.0 g (2.0 g of Rongalit) - Purity: 98 % (98 % of Purity) - Wate: 400 g (400 g of Wate) - Acrylonitrile: 34 ml (34 ml of Acrylonitrile) - Vazo: 128 mg (128 mg of Vazo) - Dimethylacetamide: 5 ml (5 ml of Dimethylacetamide) - Acrylonitrile: 35.6 g (35.6 g of Acrylonitrile)

Relevant Experimental Evidence - Example EXAMPLE: Autoclave: 5 L, Dispersion: 50 %, Fe: 0.986 g, Rongalit: 2.0 g, Purity: 98 %, Wate: 400 g, Acrylonitrile: 34 ml, Initiator: 95 ml, Pressure: 71 ml, Time: 22 hours - Example EXAMPLE: Pressure: 3 bar, Vazo: 128 mg, Dimethylacetamide: 5 ml, Time: 6 hours, Acrylonitrile: 35.6 g - Example EXAMPLE 6: Conversion: 39 %, Conversion: 48 %, Dmac: 2 g, Conversion: 9.2 % - Example EXAMPLE: - Example EXAMPLE: - Example EXAMPLE:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US8470422B2
- Patent Title: Nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013-06-25
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Elongation: 400 % (400 % of Elongation) - Dpg: 0.5 phr (0.5 phr of Dpg) - Zmbt: 0.25 phr (0.25 phr of Zmbt) - Zdec: 0.25 phr (0.25 phr of Zdec)

Relevant Experimental Evidence - Example EXAMPLE 12: Elongation: 400 %, Dpg: 0.5 phr, Zmbt: 0.25 phr, Zdec: 0.25 phr

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: EP3902855B1
- Patent Title: Pulverulent mixture of precrosslinked nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2026-05-06
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example EXAMPLE:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9023914B2 is the only patent in the set that clearly discloses solution polymerization in an organic solvent; it uses free-radical polymerization with specific modifier substances and optional hydrogenation, whereas US8470422B2 and EP3902855B1 are centered on latex/article formulation and powder processing, respectively.
- US9023914B2 reports batchwise preparation in a 5 L autoclave, with butadiene metered under 3 bar and reactions started at about 100 °C; example conversions range from 9.2% to 56% over 6 to 22 hours. The other two patents do not provide comparable reactor pressure/time/conversion data in the supplied extraction.
- US8470422B2 defines the high-acrylonitrile latex blend by composition: total acrylonitrile 17–45 wt%, methacrylic acid <15 wt%, butadiene remainder, and a first/second terpolymer ratio of 60:40 to 40:60. US9023914B2 and EP3902855B1 do not disclose these blend ratios or carboxylated latex targets.
- EP3902855B1 is the only patent that explicitly requires a polyfunctional crosslinker, trimethylolpropane tri(meth)acrylate, at 0.5–5 wt% of the overall polymer, together with separating agents for powder formation; the other patents instead emphasize modifier chemistry in solution (US9023914B2) or ionic/sulfur cure balance in latex articles (US8470422B2).

- US8470422B2 emphasizes dual crosslinking in the latex state: ionic crosslinking via alkali/metal oxide and covalent sulfur/accelerator cure, with pH 9–12.5 during glove formation; this is a formulation/process strategy distinct from the RAFT-like modifier strategy in US9023914B2 and the precrosslinked powder strategy in EP3902855B1.
- A recurring compositional motif across all three patents is the use of acrylonitrile with butadiene-based backbones, but the newer claims diversify the product form: solution-synthesized specialty NBR/HNBR (US9023914B2), soft glove latex blends (US8470422B2), and precrosslinked powder rubbers for compounding (EP3902855B1).

4. CONCLUSION

The evidence indicates that high-acrylonitrile NBR development is being driven by three complementary levers: solution-phase molecular architecture control, latex blending for article softness, and precrosslinked powder design for compounding performance. For synthesis of specialty high-acrylonitrile NBR, US9023914B2 supports free-radical solution polymerization in organic solvent at about 100 °C in a pressurized batch reactor, with modifier chemistry used to control molecular weight and polydispersity and with optional hydrogenation downstream. For glove-type carboxylated NBR, US8470422B2 suggests that a 17–45 wt% acrylonitrile window, methacrylic acid below 15 wt%, and a roughly 50:50 blend of two terpolymer grades can reduce modulus while preserving strength when ionic and sulfur cure are balanced. For powder applications, EP3902855B1 supports precrosslinking with 0.5–5 wt% TMPTMA and use of separating agents to obtain processable nitrile rubber powders. Overall, the strongest synthesis guidance from the set is to match the polymerization route and crosslink strategy to the end use: solution polymerization for controlled specialty NBR/HNBR, latex blending for soft elastomeric articles, and controlled precrosslinking for powder handling and downstream mixing.

5. REFERENCES

- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
- US8470422B2 | Nitrile rubber article having natural rubber characteristics | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8470422B2/en>
- EP3902855B1 | Pulverulent mixture of precrosslinked nitrile rubbers | Not disclosed | EP | 2026 | <https://patents.google.com/patent/EP3902855B1/en>